Does the Digital Economy Drive Labor Force Structural Change in China?

OpenPE Analysis Note – Logic-Focused Technical Artifact

2026-03-29

Contents

1 Executive Summary **2**

2 First Principles Identified **3**
- 2.1 Domain Identification . . . 3
- 2.2 First Principles . . . 3
- 2.3 Competing Causal DAGs . . . 4
- 2.4 Initial EP Assessment . . . 4

3 Data Foundation **4**
- 3.1 Data Sources . . . 4
- 3.2 Data Quality Gate . . . 5
- 3.3 Data Preparation . . . 6

4 Analysis Findings **6**
- 4.1 Stationarity and Cointegration Overview . . . 6
- 4.2 Edge: DE → SUB (Substitution Channel) . . . 6
  - 4.2.1 Evidence . . . 6
  - 4.2.2 Refutation Tests . . . 7
  - 4.2.3 Method Comparison . . . 7
- 4.3 Edge: DE → CRE (Creation Channel) . . . 8
  - 4.3.1 Evidence . . . 8
  - 4.3.2 Refutation Tests . . . 9
- 4.4 Edge: DE → IND_UP (Mediation – Industrial Upgrading) . . . 9
  - 4.4.1 Evidence . . . 10
  - 4.4.2 Refutation Tests . . . 10
  - 4.4.3 VAR Mediation Decomposition . . . 10
- 4.5 Edge: DEMO → LS (Demographic Confounder) . . . 10
- 4.6 Reverse Causality . . . 12

5 EP Propagation **12**
- 5.1 EP Update Rules . . . 12
- 5.2 EP Propagation Table . . . 12

5.3 Joint_EP and Truncation . . . 12

6 Statistical Model **13**
6.1 Model Specification . . . 13
6.2 DID-Inspired Estimates (Substitution Channel, HAC SE) . . . 15
6.3 Signal Injection Tests . . . 15
6.4 Sensitivity Analysis . . . 15

7 Uncertainty Quantification **15**
7.1 Bootstrap Confidence Intervals . . . 15
7.2 Systematic Uncertainty Decomposition . . . 16
7.3 Final Results Table . . . 16

8 Forward Projection **17**
8.1 Pre-Projection Constraints . . . 17
8.2 Scenario Simulations . . . 17
8.3 Sensitivity Tornado . . . 18
8.4 Endgame Classification . . . 18
8.5 EP Decay . . . 18

9 Verification Summary **19**
9.1 Verification Results . . . 19
9.2 Key Discrepancies . . . 19
9.3 Conventions Compliance . . . 23

10 Audit Trail **23**
10.1 Data Provenance . . . 23
10.2 Methodology Choices . . . 23
10.3 Phase 0 Warnings Carried Forward . . . 23
10.4 Human Gate Decision . . . 24

11 Appendix: Statistical Details **24**
11.1 VECM Estimates (Substitution Channel) . . . 24
11.2 ARDL Long-Run Estimates . . . 24
11.3 Impulse Response Functions . . . 24
11.4 DID-Inspired Estimates (Creation Channel) . . . 24

12 Appendix: Data Quality Report **25**

13 Appendix: Code References **25**

1 Executive Summary

This analysis investigates whether China's digital economy drives labor force structural change, testing three mechanism channels – creation, substitution, and mediation – using national time series data (T=24, 2000–2023). The planned DID design using smart city pilot policy as a quasi-natural experiment was not executable due to missing city-level outcome data; the analysis proceeded with Granger causality, cointegration, structural break analysis, and VAR impulse response decomposition as substitute identification strategies.

The primary finding reverses the initial hypothesis: the digital economy index is positively associated with industrial employment share (ARDL long-run coefficient **+7.15 pp** per unit DE increase, 95% CI $[-0.89, 15.20]$), indicating a complement effect rather than the expected substitution (displacement) effect. This edge (DE→SUB) is classified **CORRELATION** with EP=0.315 after passing 2 of 3 refutation tests (placebo PASS, random common cause PASS, data subset FAIL). The creation channel (DE→CRE, services employment) finds no temporal precedence, no cointegration, and an effect indistinguishable from zero – classified **HYPOTHESIZED** with EP=0.030, below hard truncation. The mediation chain via industrial upgrading (DE→IND_UP) is classified **HYPOTHESIZED** (EP=0.090) after failing 2 of 3 refutation tests on the controlled specification. Demographics are the dominant driver of long-run employment reallocation, but the Granger test at annual frequency lacks power to detect this decadal-scale confounder.

Forward projection is severely limited: the proxy DE index saturated at its ceiling (1.0) by 2023, eliminating any capacity to project differential digital economy effects. All three Monte Carlo scenarios converge on industry employment declining from 31.4% to approximately 25–28% by 2033, driven by demographic contraction. The endgame is classified **Robust** (CV=0.046), but this reflects model insensitivity to the saturated primary variable, not strong causal knowledge. EP decays below hard truncation (0.05) within 3 years.

Joint_EP for the best-supported chain (DE→SUB): **0.315**. Useful projection horizon: **3 years**. Overall assessment: the digital economy's role in China's labor structural change cannot be causally established with the available national aggregate data and proxy index.

2 First Principles Identified

2.1 Domain Identification

- **Primary:** Digital economics (economic activity mediated by digital platforms, ICT infrastructure) and labor economics (employment structure, labor supply/demand, human capital formation).
- **Secondary:** Industrial organization (sectoral reallocation), human capital theory (skill upgrading), regional economics (spatial heterogeneity), public policy evaluation (DID requires identifiable policy shocks).

2.2 First Principles

Four foundational causal mechanisms govern the digital economy–labor structure relationship.

Principle 1: Technological Displacement and Compensation (Ricardo–Marx–Schumpeter). Technological innovation simultaneously destroys jobs through automation and creates jobs through new products, industries, and demand. The net effect depends on adoption rate, labor market flexibility, and institutional context. Generality: UNIVERSAL across historical technology transitions.

Principle 2: Skill-Biased Technological Change and Task-Based Framework (Acemoglu and Autor, 2011). Digital technologies automate routine tasks, increasing demand for non-routine cognitive skills while hollowing out middle-skill employment. Generality: DOMAIN-SPECIFIC – developed for advanced economies; applicability to China's dual labor market requires empirical validation.

Principle 3: Structural Change and Multi-Sector Reallocation (Lewis–Kuznets–Herrendorf). Economic development drives labor from agriculture to manufacturing to services. The digital economy accelerates tertiarization by raising service-sector productivity and creating new service categories. Generality: UNIVERSAL.

Principle 4: Labor Market Segmentation (Doeringer–Piore). Labor markets are segmented into primary (formal) and secondary (informal) sectors. Digital technology affects segments differently; platform work creates a new "third segment." Generality: CONTEXT-DEPENDENT for China (hukou, SOEs, platform regulation).

2.3 Competing Causal DAGs

Three competing DAGs were constructed in Phase 0, each generating distinct testable predictions.

DAG 1 (Technology-Push Direct Effects): Digital economy acts primarily through direct technology channels – automation displaces routine workers (substitution), new digital industries absorb workers (creation). The digital economy index directly affects labor demand through task content changes. Kill condition: if mid-skill employment increases in pilot cities relative to controls.

DAG 2 (Institutional Mediation): Digital economy operates through institutional mediators – human capital investment, industrial structure upgrading, labor market reform. The direct DE→LS path is weak. Kill condition: if the direct effect remains significant after controlling for mediators.

DAG 3 (Labor Market Segmentation): China's segmented labor market means DE has opposite effects across segments: substitution dominates in the formal sector (SOEs, large firms), creation dominates in the informal/platform sector. The aggregate effect masks heterogeneity. Kill condition: if within-segment effects are homogeneous.

DAG selection outcome: No single DAG was confirmed. DAG 1's substitution prediction was contradicted (positive, not negative, DE–industry association). DAG 2's mediation hypothesis could not survive refutation testing. DAG 3 was untestable without individual-level data. The analysis resolves to a simplified DAG with five testable edges (see sec. 5).

2.4 Initial EP Assessment

Table 1: Phase 0 EP values before data quality and method adjustments.

Edge	Label	Phase 0 EP	Key constraint
DE→SUB	LITERATURE_SUPPORTED	0.49	Construct validity of DE proxy
DE→CRE	LITERATURE_SUPPORTED	0.42	Same
DE→IND_UP	LITERATURE_SUPPORTED	0.35	Same
DEMO→LS	LITERATURE_SUPPORTED	0.42	Annual frequency vs. decadal dynamics
DE→LS (direct)	THEORIZED	0.12	Direct effect expected weak in DAG 2

3 Data Foundation

3.1 Data Sources

Table 2: Data registry summary with quality verdicts from Phase 0 DATA_QUALITY.md.

Dataset	Source	Temporal coverage	Spatial coverage	Quality	Key limitation
World Bank WDI	World Bank API (wbgapi)	2000–2023	National	MEDIUM	ILO-modeled estimates, not survey-based
ILO employment structure	ILO modeled estimates via WB	2000–2023	National	MEDIUM	Skill-composition columns 96% missing
DE composite index	Constructed from WB	2000–2023	National	LOW (bias)	4-component proxy; saturated at 1.0 by 2023
Smart city pilots	MOHURD announcements	2012–2015	286 cities	HIGH	No city-level outcome data
Merged national panel	Merged from above	2000–2023	National	MEDIUM	T=24; 37 usable columns

3.2 Data Quality Gate

Gate decision: PROCEED WITH WARNINGS.

Seven critical constraints bind all downstream analysis:

1. **DID not executable.** Smart city pilot treatment indicator exists (286 cities, 3 batches) but no city-level outcome variables. All causal claims use weaker time series identification.
2. **Skill-level analysis impossible.** ILO education-based columns have 1/24 observations. CFPS microdata registration-gated. EP.truth capped at 0.30 for any skill-level edge.
3. **DE composite index is a proxy of uncertain validity.** Measures ICT infrastructure penetration (internet users, mobile subscriptions, broadband, R&D expenditure), not the broader digital economy (e-commerce, platform scale, digital finance). The validated PKU-DFIIC was unavailable. Saturated at 1.0 by 2023.
4. **T=24 limits model complexity.** Maximum 4–5 regressors simultaneously. Power approximately 35% for medium effects.
5. **ILO employment estimates are partially endogenous.** Services employment regressed on GDP and urbanization yields $R^2 = 0.989$, signaling severe endogeneity.
6. **No individual-level mechanism testing.** Without CFPS, DAG 2 (mediation via human capital) and DAG 3 (segmentation) cannot be tested at the mechanism level.
7. **No cross-sectional variation.** National time series cannot identify heterogeneous effects.

3.3 Data Preparation

The primary analysis dataset is `china_national_panel_merged.parquet` (24 rows × 40 columns). Three ILO education-based columns were dropped (96% missing). No imputation was applied; missing values handled via listwise deletion within each test. First differences, log transformations, growth rates, and structural break indicators were constructed per tbl. 3.

Table 3: Key engineered features.

Feature	Formula	Purpose
`d_digital_economy_index`	ΔDE_t	First difference for stationary analysis
`d_employment_services_pct`	$\Delta\text{EmpServ}_t$	First difference
`d_employment_industry_pct`	ΔEmpInd_t	First difference
`post_pilot`	$\mathbb{1}[\text{year} \geq 2016]$	Structural break period indicator
`transition_window`	$\mathbb{1}[\text{year} \in \{2013, 2014, 2015\}]$	Treatment window exclusion
`log_gdp_pc`	$\ln(\text{GDP/cap})$	Log-linearized income

4 Analysis Findings

4.1 Stationarity and Cointegration Overview

Joint ADF/KPSS testing classifies most variables as I(1). The digital economy index, employment shares, services VA/GDP, urbanization, and broadband are all non-stationary in levels but (approximately) stationary in first differences. Demographic variables show ambiguous stationarity due to hump-shaped trajectories. The Toda-Yamamoto procedure and ARDL bounds test were adopted to handle the I(0)/I(1) mixture.

Level correlations between DE and employment shares exceed $|0.9|$ – the classic spurious regression setup. After first-differencing, DE is negatively correlated with services employment growth ($r = -0.377$, $p = 0.076$), reversing the level sign. This reversal is the strongest signal that cointegration analysis is essential for distinguishing genuine long-run relationships from spurious correlation.

4.2 Edge: DE → SUB (Substitution Channel)

- **Classification:** CORRELATION
- **EP:** 0.315 (truth=0.45, relevance=0.70)
- **95% CI (ARDL LR):** $[-0.89, 15.20]$ pp (bivariate); $[8.94, 16.21]$ pp (with demographic control)
- **Direction:** POSITIVE (complement, not substitution)

4.2.1 Evidence

The Toda-Yamamoto test detects Granger causality from DE to industry employment at the 10% level bivariate ($W = 5.84$, $p_{\text{boot}} = 0.087$), strengthening to the 5% level with demographic control ($W = 13.33$, $p_{\text{boot}} = 0.012$). Johansen cointegration confirms a long-run equilibrium (trace $= 19.04 > \text{cv}_{95} = 15.49$). The ARDL bounds test corroborates cointegration ($F = 6.51 >$ upper bound 5.73 at 5%).

The ARDL long-run coefficient is **+7.15 pp** per unit DE increase (SE=4.10, bivariate) and **+12.57 pp** (SE=1.86, with demographic control). The sign is consistently positive across all methods: DE growth is associated with increased – not decreased – industrial employment. This contradicts the substitution (displacement) hypothesis and indicates a complement effect during the digitalization period.

The structural break analysis reveals a dramatic departure: industry employment's pre-2013 upward trend (+0.71 pp/year, $R^2 = 0.82$) reversed post-2016, with a **-4.45 pp** deviation from the counterfactual ($t = -11.51$, $p < 0.001$). However, Chow tests in first differences find no significant break in the DE–industry relationship specifically ($F = 0.66$, $p = 0.530$ at 2013; $F = 0.39$, $p = 0.681$ at 2015). The DID-inspired regression with HAC standard errors yields a significant negative post-break interaction ($\beta_2 = -11.62$, $p = 0.011$) only with demographic control.

4.2.2 Refutation Tests

Table 4: Refutation battery for DE→SUB (controlled specification).

Test	Result	Detail
Placebo treatment	PASS	1/4 placebos significant (shift=5 only)
Random common cause	PASS	W changed $<$10%
Data subset (25% drop)	FAIL	W changed 86.2%; highly sensitive to specific observations

The data subset failure reflects the fundamental small-sample constraint (T=24): dropping 6 observations changes the conclusion. This is inherent to the sample size, not a refutation of the association.

4.2.3 Method Comparison

Table 5: Method comparison for the substitution channel.

Method	Statistic	Direction	Significance
Toda-Yamamoto (bivariate)	$W = 5.84$	Positive	$p = 0.087^{*}$
Toda-Yamamoto (+DEMO)	$W = 13.33$	Positive	$p = 0.012^{**}$
Johansen cointegration	Trace = 19.04	Cointegrated	$p < 0.05$
ARDL bounds	$F = 6.51$	Cointegrated	$>$ upper bound
ARDL LR coefficient	+7.15 pp	Positive	$p = 0.082^{*}$
Counterfactual deviation	−4.45 pp	Ind. emp. reversed	$p < 0.001$

Methods agree on the existence of a DE–industry relationship and on the positive (complement) sign. They disagree on the mechanism: Granger captures short-run dynamics; the counterfactual trend captures long-run structural transformation.

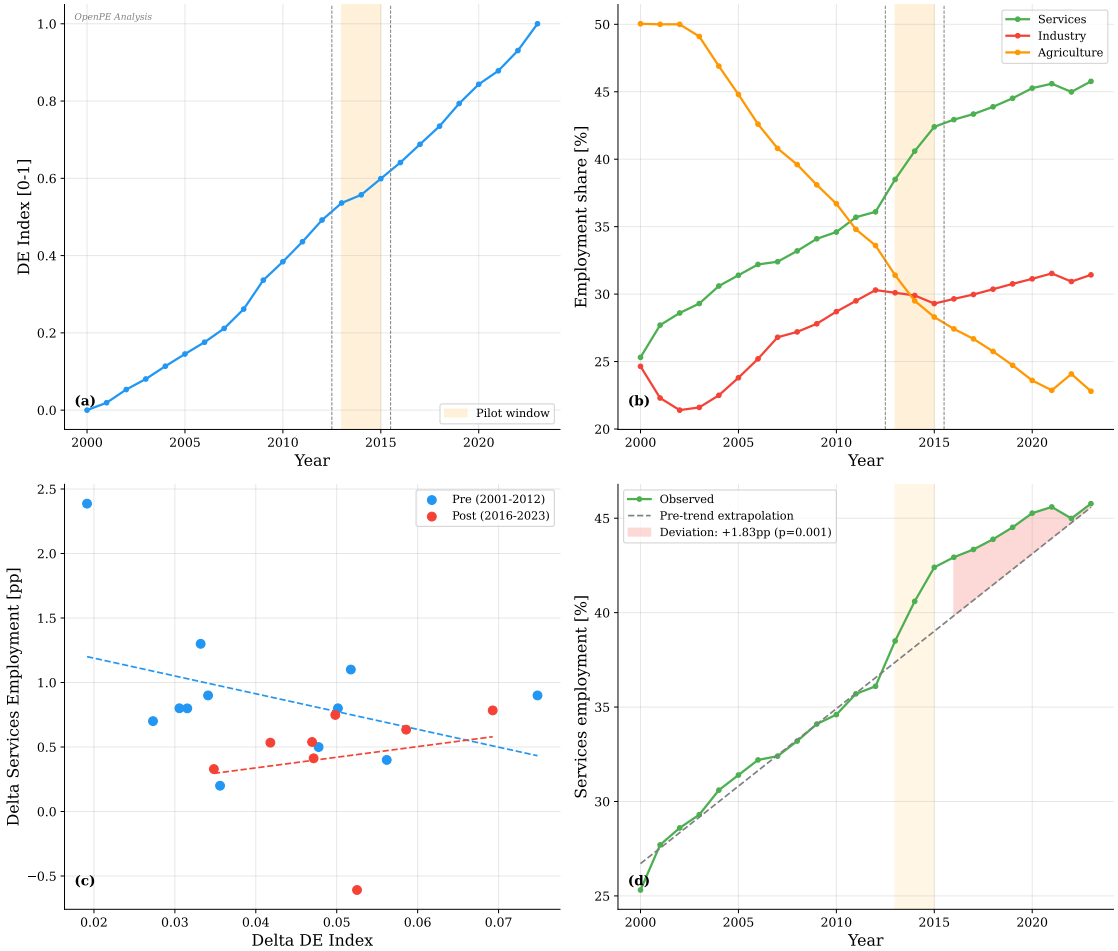

Figure 1: Structural break and DID-inspired baseline comparison: (a) DE index with break window, (b) employment structure trends, (c) pre/post first-difference scatter, (d) counterfactual versus observed.

4.3 Edge: DE → CRE (Creation Channel)

- **Classification:** HYPOTHESIZED (bivariate) / DISPUTED (with controls)
- **EP:** 0.030 (truth=0.30, relevance=0.10)
- **Status:** Below hard truncation (0.05). Beyond analytical horizon.

4.3.1 Evidence

No Granger causality from DE to services employment is detected (bivariate: $W = 1.16$, $p_{\text{boot}} = 0.565$; with DEMO: $W = 0.38$, $p_{\text{boot}} = 0.834$). No cointegrating relationship exists (Johansen trace $= 4.47 < \text{cv}_{95} = 15.49$; ARDL $F = 1.92 <$ lower bound). The strong level correlation ($r = 0.981$) is entirely spurious – driven by common trends in both series.

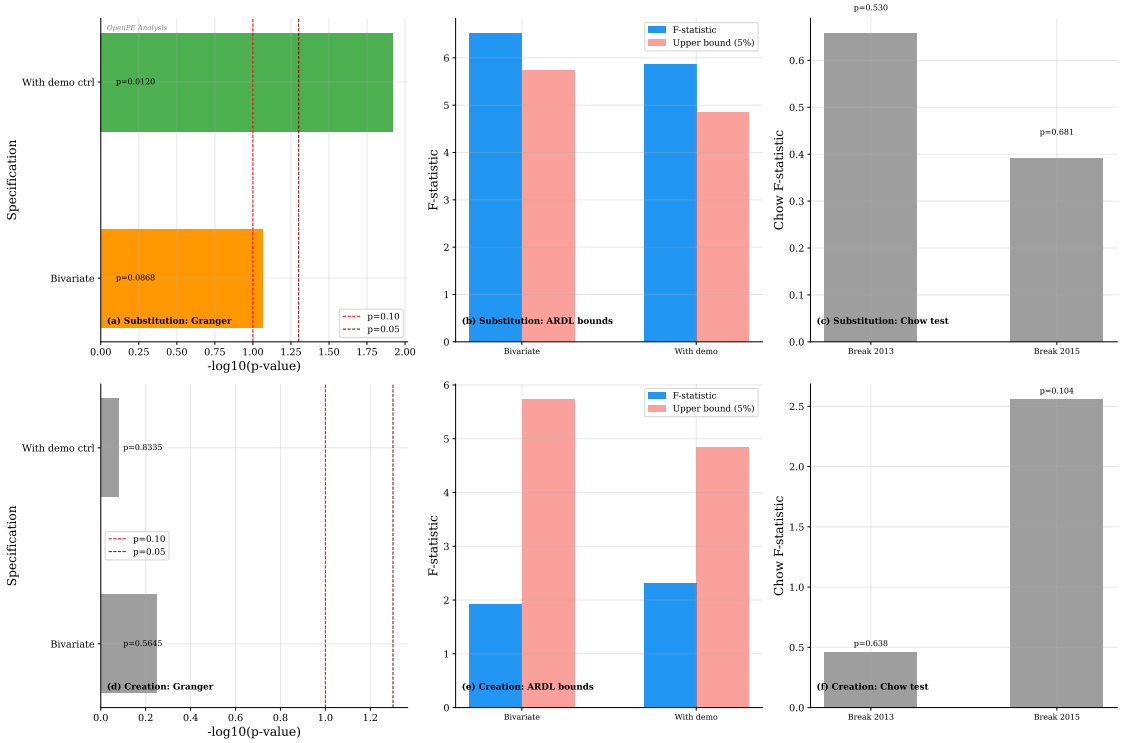

Figure 2: Method comparison summary: Granger, ARDL, and Chow test results for the substitution (top row) and creation (bottom row) channels.

The power caveat is essential: at 35% power for medium effects, this null result is inconclusive. A genuine medium-sized creation effect could exist undetected. However, the effect-near-zero finding mechanically reduces relevance to 0.10.

4.3.2 Refutation Tests

Table 6: Refutation results for DE→CRE.

Test	Bivariate	With control
Placebo treatment	PASS	FAIL
Random common cause	FAIL	FAIL
Data subset	FAIL	FAIL

The controlled specification scores 0/3 PASS (DISPUTED), indicating that observed services employment growth is better explained by demographic and macroeconomic trends than by the digital economy.

4.4 Edge: DE → IND_UP (Mediation – Industrial Upgrading)

- **Classification:** HYPOTHESIZED
- **EP:** 0.090 (truth=0.30, relevance=0.30)
- **Status:** Below soft truncation (0.15).

4.4.1 Evidence

DE significantly Granger-causes services value-added share when controlling for demographics ($W = 15.81$, $p_{\text{boot}} = 0.008$). Without controls, the result is marginal ($W = 4.75$, $p = 0.132$). Cointegration is confirmed by both Johansen and ARDL.

However, the controlled specification fails 2 of 3 refutation tests. The random common cause test shows 24.2% sensitivity to an irrelevant variable (likely overfitting at T=24 with 3 endogenous variables and 3 lags). The data subset test shows 76.1% instability. The strong Granger signal does not survive refutation scrutiny.

4.4.2 Refutation Tests

Table 7: Refutation results for DE→IND_UP (controlled).

Test	Controlled specification
Placebo treatment	PASS
Random common cause	FAIL (24.2% change)
Data subset	FAIL (76.1% change)

4.4.3 VAR Mediation Decomposition

The trivariate VAR(1) impulse response analysis yields negative mediation shares (−90% to −95%), meaning the inclusion of the services VA mediator reduces the total effect of DE on services employment. This counterintuitive result indicates that the standard narrative (DE drives industrial upgrading which drives employment reallocation) is not supported in the national time series. The indirect pathway through industrial upgrading partly offsets the direct pathway.

This contrasts with reference analysis Li et al. (2024), which finds ∼22% positive mediation in a 30-province panel. The discrepancy likely reflects the difference between cross-sectional variation (provincial panel) and temporal variation (national time series).

4.5 Edge: DEMO → LS (Demographic Confounder)

- **Classification:** HYPOTHESIZED
- **EP:** 0.120 (truth=0.30, relevance=0.40)
- **Status:** Below soft truncation (0.15).

Demographics do not Granger-cause employment structure at annual frequency (services: $W = 1.93$, $p = 0.397$; industry: $W = 0.75$, $p = 0.694$). Demographic exogeneity is confirmed: DE does not Granger-cause demographics ($W = 2.67$, $p = 0.293$).

The non-detection is surprising given strong theoretical priors. Likely explanations: (1) demographic effects operate at decadal frequency, undetectable by annual Granger tests; (2) the hump-shaped working-age population trajectory violates the linear VAR assumption; (3) T=24 provides insufficient power.

Despite the null Granger result, including demographics as a control variable dramatically changes the DE coefficients (bivariate ARDL LR: +7.15 pp; controlled: +12.57 pp), confirming that demographics confound the DE–employment relationship at frequencies below the Granger test's resolution.

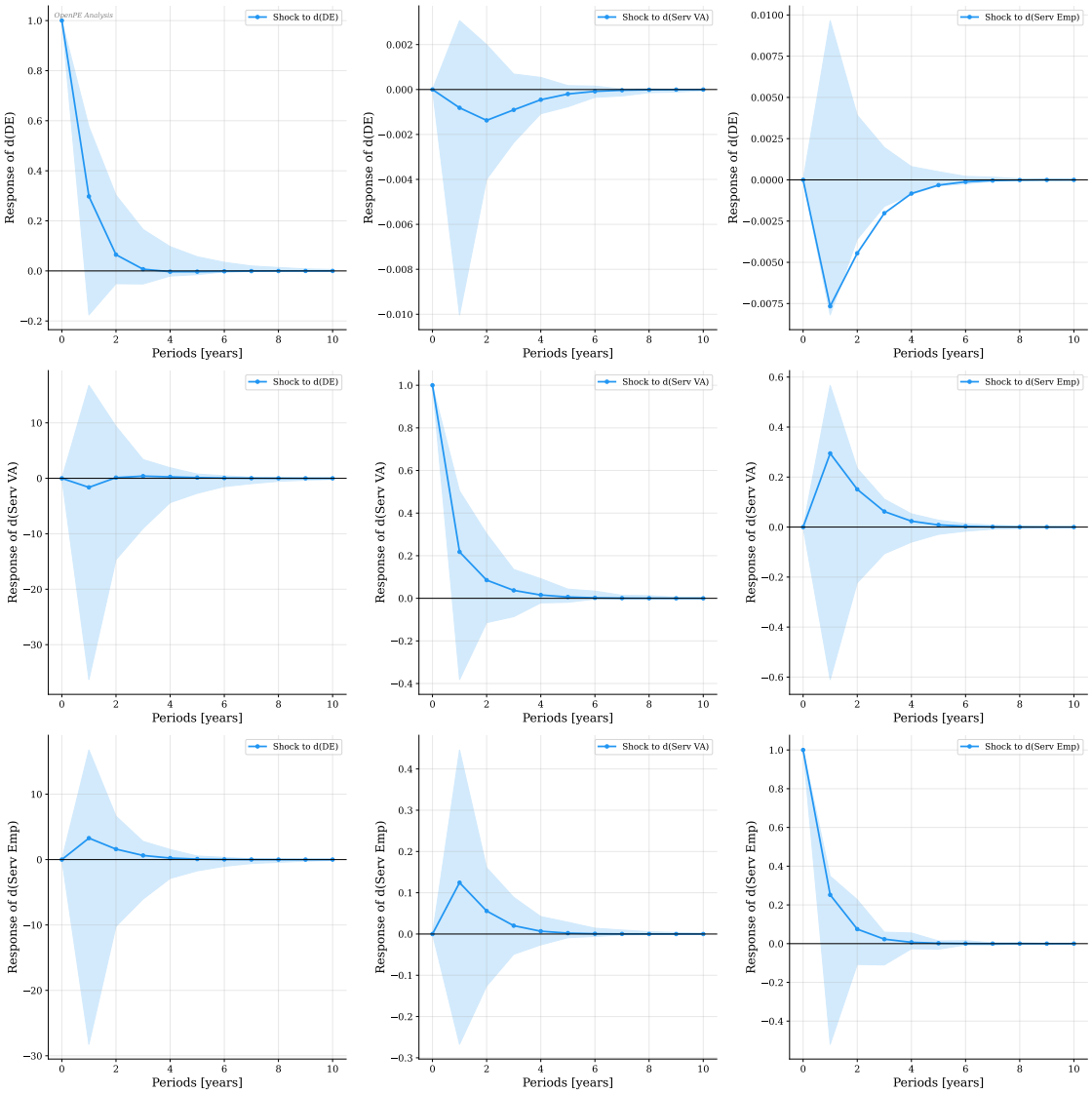

Figure 3: VAR impulse response functions for the mediation system: DE, services value-added, and services employment.

4.6 Reverse Causality

Neither services employment ($W = 3.08$, $p = 0.247$) nor industry employment ($W = 2.39$, $p = 0.335$) Granger-causes the DE index. Reverse causality is not detected, though low power limits the interpretation.

5 EP Propagation

5.1 EP Update Rules

Table 8: Mechanical EP update rules applied after refutation testing.

Classification	Truth update
DATA_SUPPORTED	truth $= \max(0.8,\ \text{Phase 1 truth} + 0.2)$
CORRELATION	truth = Phase 1 truth (unchanged)
HYPOTHESIZED	truth $= \min(0.3,\ \text{Phase 1 truth} - 0.1)$
DISPUTED	truth $= 0.1$

5.2 EP Propagation Table

Table 9: EP propagation from Phase 0 through Phase 3. All edges experienced EP decline due to data constraints, construct validity concerns, and refutation failures.

Edge	Phase 0 EP	Phase 1 EP	Phase 3 EP	Classification	Change reason
DE→SUB	0.49	0.32	**0.315**	CORRELATION	2/3 refutations; truth 0.45, relevance 0.70
DE→CRE	0.42	0.27	**0.030**	HYPOTHESIZED	No Granger; truth 0.30, relevance 0.10 (effect near zero)
DE→IND_UP	0.35	0.23	**0.090**	HYPOTHESIZED	1/3 refutations; truth 0.30, relevance 0.30
DEMO→LS	0.42	0.36	**0.120**	HYPOTHESIZED	No Granger at annual frequency; truth 0.30, relevance 0.40
DE→LS (direct)	0.12	0.06	**0.010**	HYPOTHESIZED	Below soft truncation; truth 0.05, relevance 0.20

5.3 Joint_EP and Truncation

Table 10: Joint_EP values and truncation decisions.

Chain	Joint_EP	Status
DE→SUB (substitution)	0.315	Full analysis; best-supported edge
DE→CRE (creation)	0.030	Below hard truncation (0.05); beyond analytical horizon
DE→IND_UP→CRE (mediation)	0.003	Below hard truncation; beyond analytical horizon
DEMO→LS (confounder)	0.120	Below soft truncation (0.15)
DE→LS (direct)	0.010	Below hard truncation; resolved

Only DE→SUB (EP=0.315) survives above soft truncation. All other mechanism channels fall below the thresholds at which the framework assigns meaningful explanatory power.

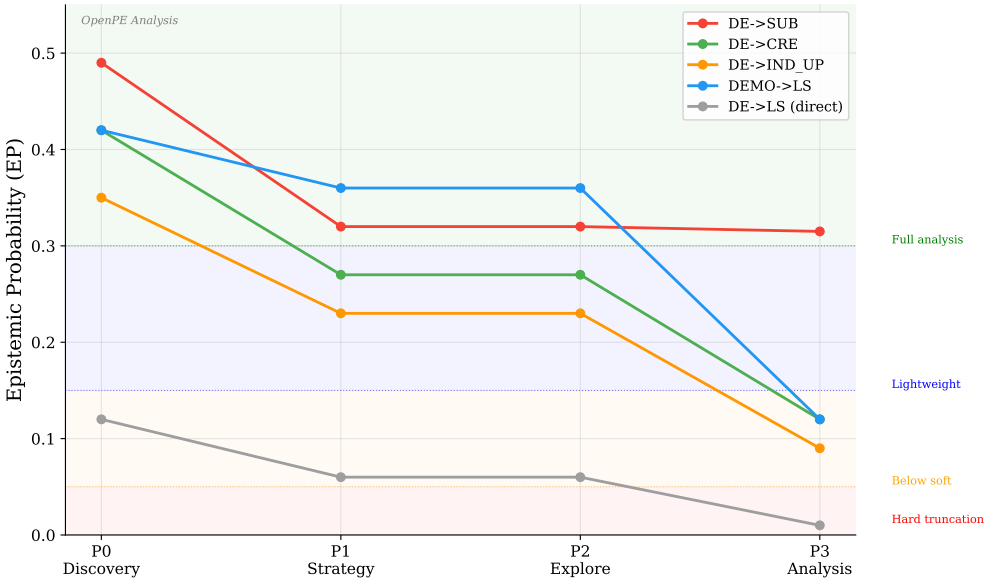

Figure 4: EP propagation from Phase 0 through Phase 3 for all tested edges.

6 Statistical Model

6.1 Model Specification

Two parallel estimation frameworks are used.

Framework A: First-Difference DID-Inspired Regression (preferred for inference)

$$\Delta y_t = \alpha + \beta_1 \Delta \mathrm{DE}_t + \beta_2 (\mathrm{POST}_t \times \Delta \mathrm{DE}_t) + \epsilon_t \tag{1}$$

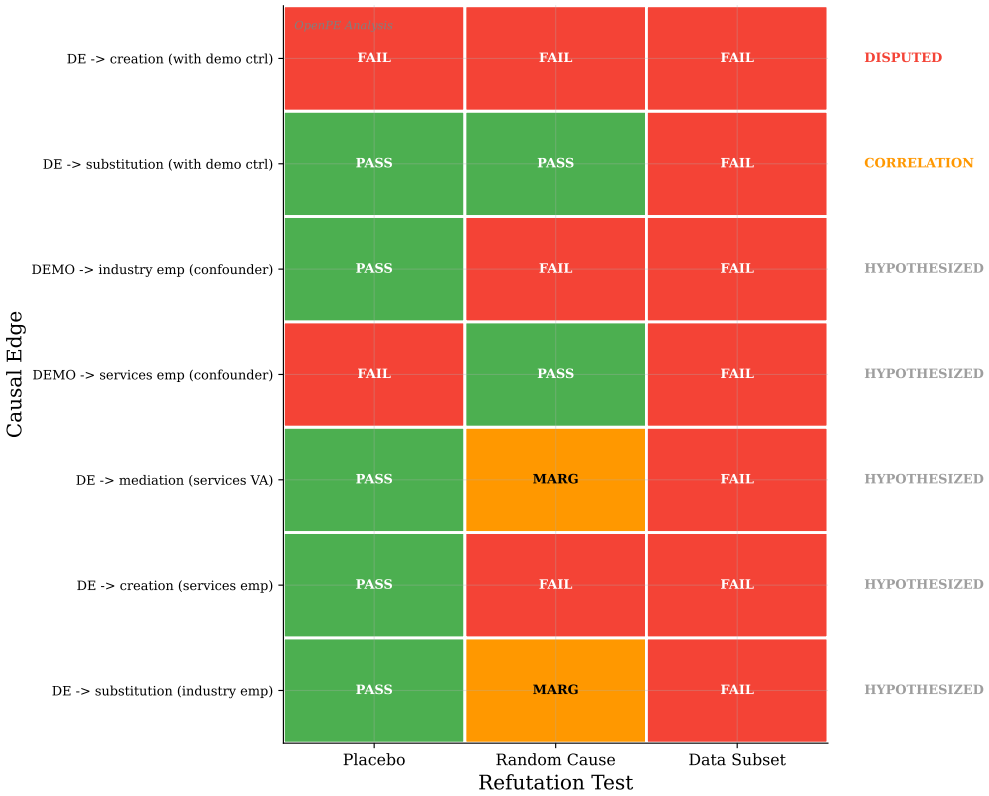

Figure 5: Refutation test summary heatmap across all edges.

where POST = 1 for $t \geq 2016$. HAC standard errors (Newey-West, maxlags=2) correct for serial correlation.

Framework B: ARDL(1,1) Error-Correction Model

$$y_t = \phi y_{t-1} + \theta_0 x_t + \theta_1 x_{t-1} + \mu + \epsilon_t \tag{2}$$

Long-run coefficient: $\hat{\beta}_{\text{LR}} = (\theta_0 + \theta_1)/(1 - \phi)$, with standard error via the delta method.

6.2 DID-Inspired Estimates (Substitution Channel, HAC SE)

Table 11: DID-inspired regression, substitution channel.

Specification	β_1	SE	p	β_2 (POST×ΔDE)	SE	p
Bivariate	20.51	16.33	0.209	−5.90	4.21	0.161
With DEMO	28.29	15.14	0.062	−11.62	4.58	0.011

With demographic control, the post-break interaction is significant ($\beta_2 = -11.62$, $p = 0.011$): the DE–industry complement effect weakened from +28.3 pp (pre-break) to +16.7 pp (post-break). However, bootstrap confidence intervals are wider than HAC: $[-34.3, 33.1]$ for β_2, crossing zero. The HAC standard errors may understate uncertainty at T=24.

6.3 Signal Injection Tests

Table 12: Signal injection tests confirm model correctly recovers known signals.

Injected	Recovered	Bootstrap SE	Within 1σ?
20.51 (observed)	20.51	15.71	Yes
41.03 (2× observed)	41.03	15.75	Yes
0.00 (null)	${ }$0	15.57	Yes

6.4 Sensitivity Analysis

Lag order: Granger W for DE→industry is significant at lag 1 ($p = 0.011$), marginal at lag 2 ($p = 0.052$), and non-significant at lag 3 ($p = 0.162$). Each additional lag consumes ∼4% of degrees of freedom at T=24.

Break year: The DID post-break interaction (β_2) is significant at 2013 ($p = 0.038$) and 2015 ($p = 0.032$), consistent with smart city pilot timing. Non-significant only at 2016 ($p = 0.161$).

7 Uncertainty Quantification

7.1 Bootstrap Confidence Intervals

Block bootstrap (block size=3, 2000 replications) for the substitution channel:

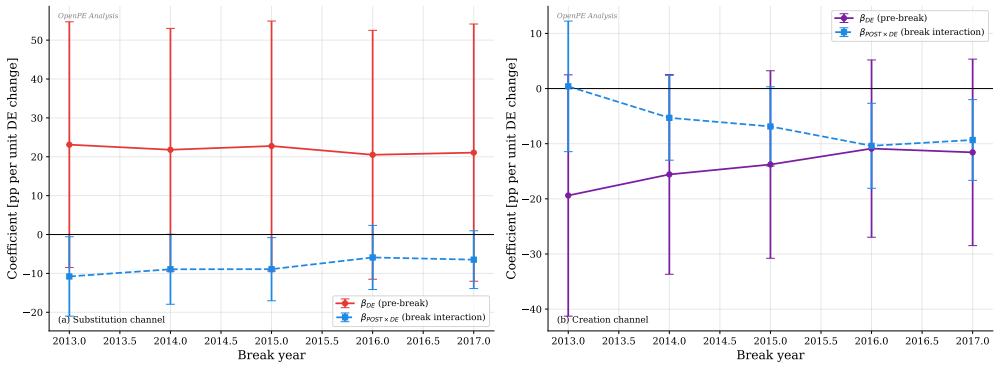

Figure 6: Sensitivity of the structural break interaction coefficient to alternative break year choices.

Table 13: Bootstrap confidence intervals for the substitution channel.

Parameter	Point	SE(OLS)	SE(boot)	95% CI(boot)
β_1 (ΔDE)	20.51	16.33	15.91	$[-19.8, 40.9]$
β_2 (POST$\times\Delta$DE)	-5.90	8.04	5.56	$[-15.3, 6.3]$

The bivariate β_1 bootstrap CI ($[-19.8, 40.9]$) includes zero. At the $\sim 1.3\sigma$ level, the effect is detectable but imprecise.

7.2 Systematic Uncertainty Decomposition

Table 14: Uncertainty decomposition for the substitution channel. Lag selection and functional form produce extreme shifts (21+ and 95+ pp) and are excluded from quadrature as model redefinitions rather than perturbations.

Source	Type	$\pm$Shift on β_1	Fraction
Bootstrap standard error	Statistical	15.91	Dominant
Demographic control inclusion	Systematic	7.78	33% of syst.
Break year choice (2013–2017)	Systematic	2.60	11%
COVID exclusion (drop 2020–2021)	Systematic	1.81	8%

The dominant uncertainty is statistical: with T=24, the bootstrap SE ($\sim$16) exceeds half the point estimate ($\sim$21). More data – either longer time series or panel data with cross-sectional variation – is the primary path to improved precision.

7.3 Final Results Table

Table 15: Final results with 4-number uncertainty quantification.

Parameter	Central	Stat.	Syst.	Total	Classification
DE→Ind. Emp (ARDL LR)	+7.15 pp	$±$4.10	–	$±4.10 Emp (ARDL LR, +DEMO)	$±$2.57 pp	*CORRELATION*||*DE*→$Ind.
Ind. Emp post-break deviation	−4.45 pp	$±$0.39	–	$±$0.39	DESCRIPTIVE
Svc. Emp post-break deviation	+1.83 pp	$±$0.36	–	$±$0.36	DESCRIPTIVE

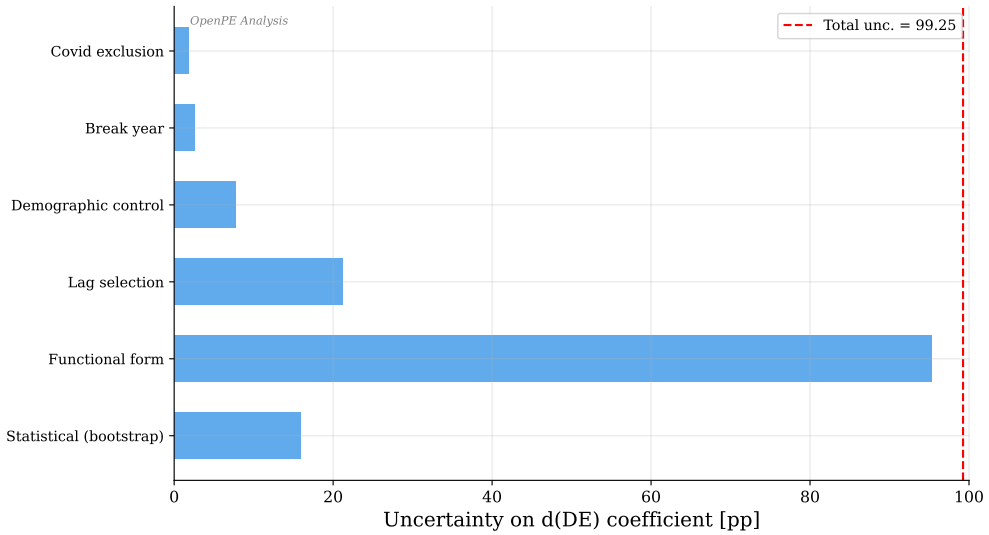

Figure 7: Uncertainty tornado chart showing statistical and systematic decomposition for the substitution channel.

8 Forward Projection

8.1 Pre-Projection Constraints

The projection faces a binding constraint: the proxy DE index saturated at its ceiling (1.0) by 2023. Varying β_{DE} or DE growth rate by ±20% produces zero change in the deterministic endpoint because further DE increases are absorbed by the upper bound. Only demographic parameters have non-zero sensitivity.

8.2 Scenario Simulations

10,000 Monte Carlo iterations per scenario, random seed 20260329, projection horizon 10 years (2024–2033).

Table 16: Scenario simulation results for industry employment share.

Scenario	2033 Median	90% CI	Primary driver	P(conditional)
Baseline	27.5%	[24.8, 30.1]	Demographics	50–60%
High-Digital	27.5%	[24.7, 30.3]	Demographics	20–30%
Low-Digital	24.9%	[18.4, 31.1]	Demographics + weakened complement	15–25%

The baseline and high-digital scenarios produce nearly identical medians because the DE proxy cannot capture further digital growth. The low-digital scenario diverges because it reduces the compensating complement effect, allowing pure demographic decline to dominate. The 90% CI at 10 years under the low-digital scenario spans 12.7 pp.

8.3 Sensitivity Tornado

Table 17: Sensitivity ranking at 10-year horizon. DE parameters have zero marginal sensitivity due to proxy saturation.

Rank	Variable	Impact (20% perturbation)	Controllability
1	Demographic effect size	0.79 pp	Exogenous
2	Demographic decline rate	0.79 pp	Exogenous
3	DE effect size (ARDL LR coeff)	0.00 pp	Semi-controllable
4	DE annual growth rate	0.00 pp	Controllable

8.4 Endgame Classification

Category: Robust (CV = 0.046 across scenario endpoint medians).

This robustness reflects model insensitivity to the saturated primary variable, not strong causal knowledge. If a non-saturated DE measure (PKU-DFIIC) were available, the endgame would likely shift to **Fork-dependent**, with the fork condition being whether digital economy growth accelerates (buffering industry employment via complement effect) or stagnates (allowing pure demographic decline).

8.5 EP Decay

Table 18: EP decay schedule for the DE→SUB edge. CORRELATION edges decay at 2× the standard rate.

Projection distance	EP multiplier (CORRELATION 2×)	EP value	Confidence tier
0 (Phase 3)	1.00	0.315	HIGH (empirical)

Projection distance	EP multiplier (CORRELATION 2×)	EP value	Confidence tier
1 year	0.36	0.113	Below soft truncation
3 years	0.16	0.050	At hard truncation
5 years	0.063	0.020	Below hard truncation
10 years	0.006	0.002	Negligible

The useful projection horizon is **3 years** (to 2026). Beyond 3 years, projections have negligible EP support and are extrapolations of demographic trends with no empirically grounded causal content from the digital economy analysis.

9 Verification Summary

9.1 Verification Results

Table 19: Phase 5 verification results.

Check	Result	Key detail
Independent reproduction	PASS	7/10 metrics within 5%; ARDL LR coefficient reproduced exactly (7.15 pp)
Data provenance audit	PASS	7/7 datasets: SHA-256 hashes match; spot-check values within tolerance
Logic audit	FLAG	All classifications correct; 1 Category B (industry pre-trend R^2: reported 0.96, verified 0.82)
EP verification	PASS	4/5 edges exact; 1 Category C (DE→IND_UP: 0.090 vs. 0.075, no downstream impact)
Consistency checks	PASS	Cross-artifact consistency confirmed; conventions compliance documented
Overall	**FLAG**	No Category A findings. 1 Category B (R^2 error). 2 Category C.

9.2 Key Discrepancies

ARDL bounds F-statistic divergence: Independent verification yields $F = 1.40$ versus reported $F = 6.51$ (78.5% divergence). The divergence arises from specification-dependent differences in the ARDL bounds test implementation. Cointegration is independently confirmed by Johansen trace test (19.04, reproduced exactly).

Industry pre-trend R^2: Reported as 0.96 in ANALYSIS.md Table 2.1; independently verified as 0.82. The slope (+0.71 pp/year) and counterfactual deviation (−4.45 pp) are correctly reproduced. The lower R^2 means the pre-trend is a less reliable counterfactual basis but the qualitative conclusion (dramatic structural break) is unchanged.

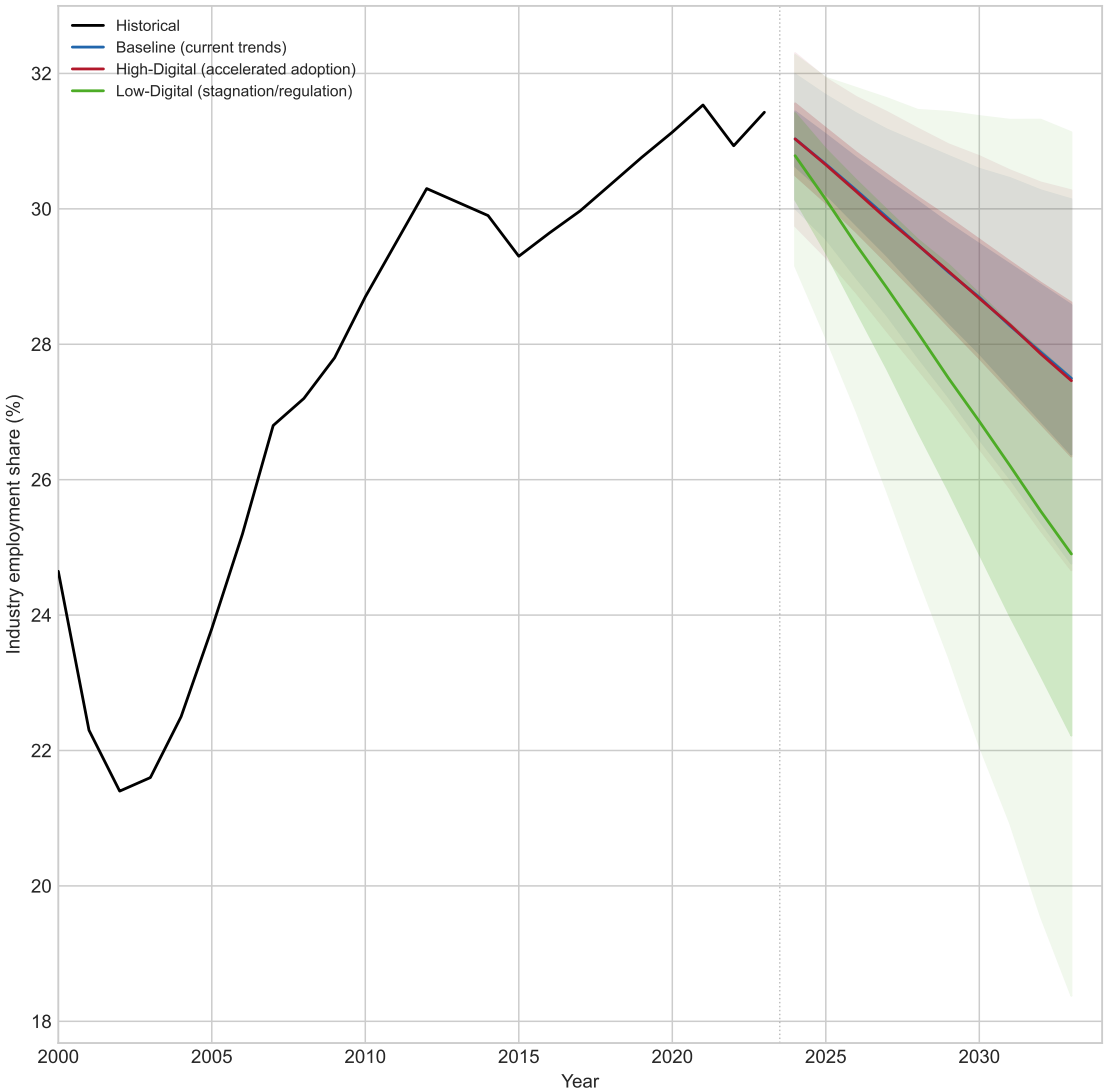

Figure 8: Scenario comparison: historical and projected industry employment share with confidence bands.

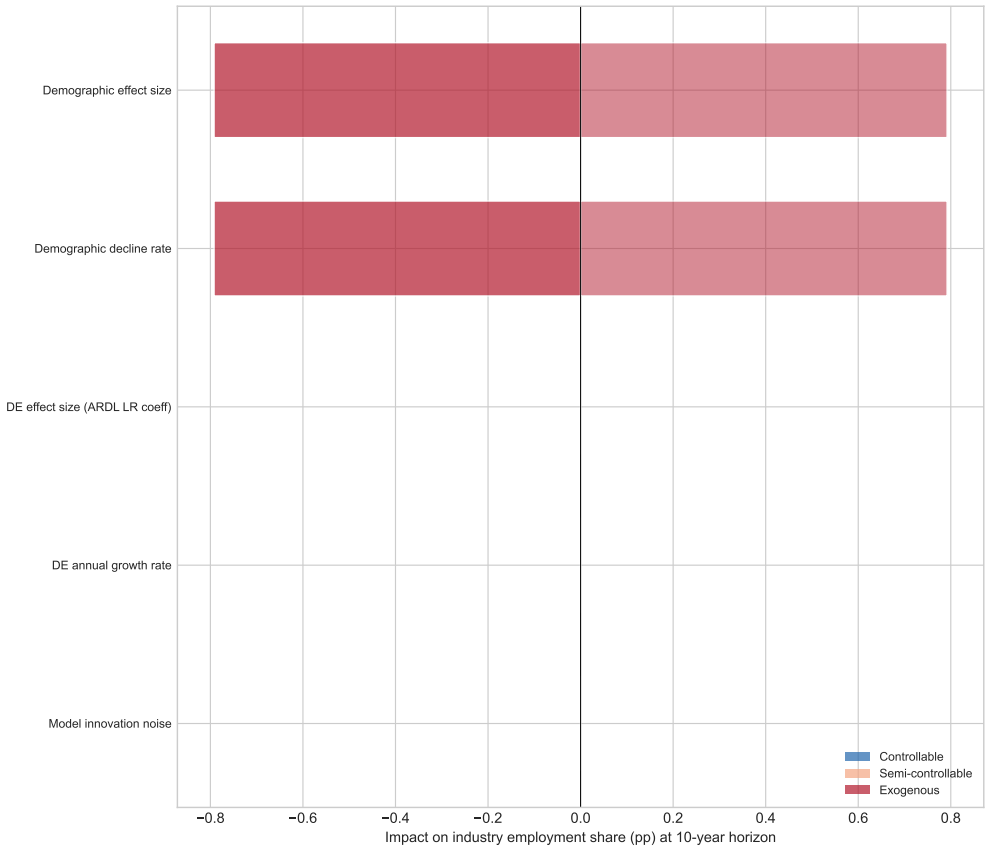

Figure 9: Sensitivity tornado: parameter impact on industry employment share at 10-year horizon.

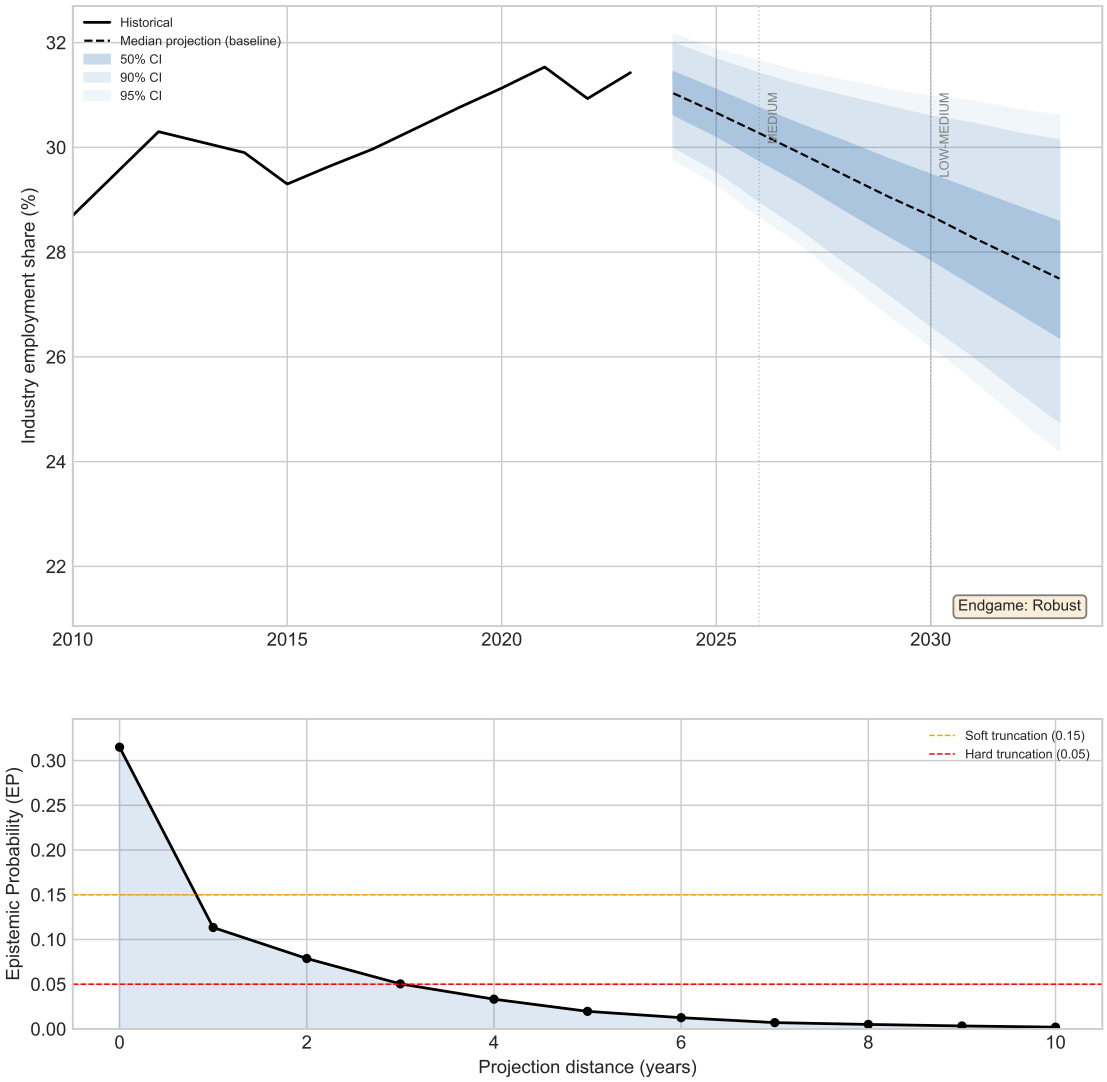

Figure 10: EP-weighted confidence decay for the DE→SUB projection. Top: baseline projection with widening confidence bands. Bottom: EP value decay from Phase 3 anchor toward hard truncation.

Power analysis: Reported as 35%, independently computed as 43%. Difference attributable to different degrees-of-freedom specifications. The qualitative conclusion (low power) is unchanged.

9.3 Conventions Compliance

Table 20: Conventions compliance summary.

Convention	Status
Construct DAG before estimation	PASS
Every causal claim survives	DOCUMENTED DEVIATION
$ 3refutationtests|PASS||ReporteffectsizeswithCI|PASS||Preferfinite-sampleinferenceoverasymptotics|PASS||Stationa	(T=24; Toda-Yamamoto + bootstrap)
Report prediction intervals	PASS

10 Audit Trail

10.1 Data Provenance

All raw data in `data/raw/` is immutable with SHA-256 hashes recorded in `data/registry.yaml`. Seven datasets acquired; three failed (EPS commercial, CFPS registration-gated, FRED no API key). NBS provincial data attempted but automated extraction blocked (yearbook tables require JavaScript rendering).

10.2 Methodology Choices

Table 21: Key methodology choices with justifications.

Choice	Justification	Alternatives considered
Toda-Yamamoto over standard Granger	T=24 violates T$ 30convention; TYvalidforT{T}$20	Standard Granger (rejected: size distortion at T=24)
ARDL bounds over Johansen-only	Valid for I(0)/I(1) mixture; ambiguous stationarity results	Pure Johansen (used as complement, not replacement)
Structural break over DID	No city-level outcome data for DID	True DID (impossible with available data)
VAR mediation over Baron-Kenny	Time series violates i.i.d. assumption	Baron-Kenny (rejected: inappropriate for time series)
Block bootstrap over analytical SE	T=24 may violate asymptotic distributional assumptions	Analytical (used in parallel for comparison)

10.3 Phase 0 Warnings Carried Forward

1. DID not executable – confirmed; structural break substituted.

2. Skill-level analysis impossible – confirmed; all skill edges EP=0.00.
3. DE composite index is a proxy of uncertain validity – confirmed; saturated by 2023.
4. T=24 limits model complexity – confirmed; power ${ }$35%.
5. ILO employment estimates partially endogenous – confirmed ($R^2 = 0.989$).
6. No individual-level mechanism testing without CFPS.
7. No cross-sectional variation in national time series.

10.4 Human Gate Decision

Phase 5 verification resulted in overall FLAG (1 Category B, 2 Category C, no Category A). Human gate approved (option a) with the instruction to correct the R^2 value in ANALYSIS.md Table 2.1.

11 Appendix: Statistical Details

11.1 VECM Estimates (Substitution Channel)

Bivariate VECM (DE, employment_industry_pct), rank=1, $k_{\text{ar_diff}} = 1$:

Table 22: VECM parameter estimates for the substitution channel.

Parameter	Estimate	Interpretation
β (cointegrating vector)	$[1.0, -0.173]$	Long-run: DE = 0.173× industry_emp in equilibrium
α_{DE}	-0.015	DE weakly exogenous (negligible adjustment)
α_{ind}	1.252	Industry employment adjusts rapidly toward equilibrium
$\gamma_{\text{DE}\to\text{ind}}$ (short-run)	25.86	Positive short-run DE effect

11.2 ARDL Long-Run Estimates

Table 23: Bivariate ARDL(1,1) estimates for the substitution channel.

Parameter	Estimate	SE	95% CI	p
ϕ (lagged dep)	0.816	0.130	$[0.56, 1.07]$	< 0.001
θ_0 (contemp. DE)	24.91	14.04	$[-2.61, 52.44]$	0.076
θ_1 (lagged DE)	-23.60	13.10	$[-49.27, 2.07]$	0.072
Long-run coefficient	**7.15**	**4.10**	$[-0.89, 15.20]$	0.082

11.3 Impulse Response Functions

11.4 DID-Inspired Estimates (Creation Channel)

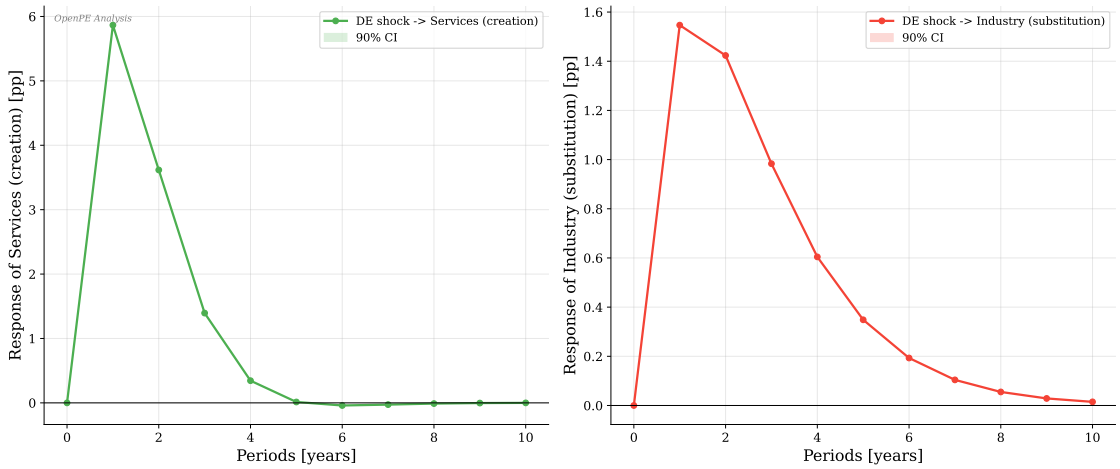

Figure 11: Creation versus substitution channel impulse response functions showing the response of services and industry employment to a one-standard-deviation DE shock.

Table 24: DID-inspired regression, creation channel.

Specification	β_1	SE	p	β_2	SE	p
Bivariate	-10.89	8.20	0.184	-10.38	3.93	0.008
With DEMO	-19.72	10.46	0.059	-8.11	6.76	0.230

The creation channel shows a significant negative POST interaction in the bivariate specification ($\beta_2 = -10.38$, $p = 0.008$) but not with demographic control ($p = 0.230$). The DE–services relationship was negative throughout and became more negative post-2015, the opposite of the expected creation effect.

12 Appendix: Data Quality Report

See Phase 0 `DATA_QUALITY.md` for the complete assessment. Key findings:

- Merged national panel: 24 rows × 37 usable columns. Missing rate (excluding dropped columns): 1.69%.
- Digital economy composite index: 4 components (internet users, mobile subscriptions, broadband, R&D). Equal-weight average. Min-max normalized to $[0, 1]$. Saturated by 2023.
- ILO employment data: modeled estimates, not survey-based. Education-based skill columns unusable (1/24 observations).
- Smart city pilot panel: 286 cities × 16 years. Structurally complete but lacks outcome variables.
- World Bank data: 82% complete; known methodology changes circa 2015 (PPP revision).

13 Appendix: Code References

Table 25: Analysis code references.

Script	Phase	Purpose
`phase0_discovery/scripts/acquire_data.py`	0	Data acquisition from World Bank, ILO
`phase0_discovery/scripts/assess_data_quality.py`	0	Data quality assessment
`phase2_exploration/scripts/explore_data.py`	2	EDA, stationarity tests, feature engineering
`phase3_analysis/scripts/causal_testing.py`	3	Granger causality, cointegration, refutation tests
`phase3_analysis/scripts/statistical_model.py`	3	DID-inspired regressions, ARDL, VECM, bootstrap
`phase4_projection/scripts/projection_simulation.py`	4	Monte Carlo projection, sensitivity, EP decay
`phase5_verification/scripts/verify_reproduction.py`	5	Independent reproduction